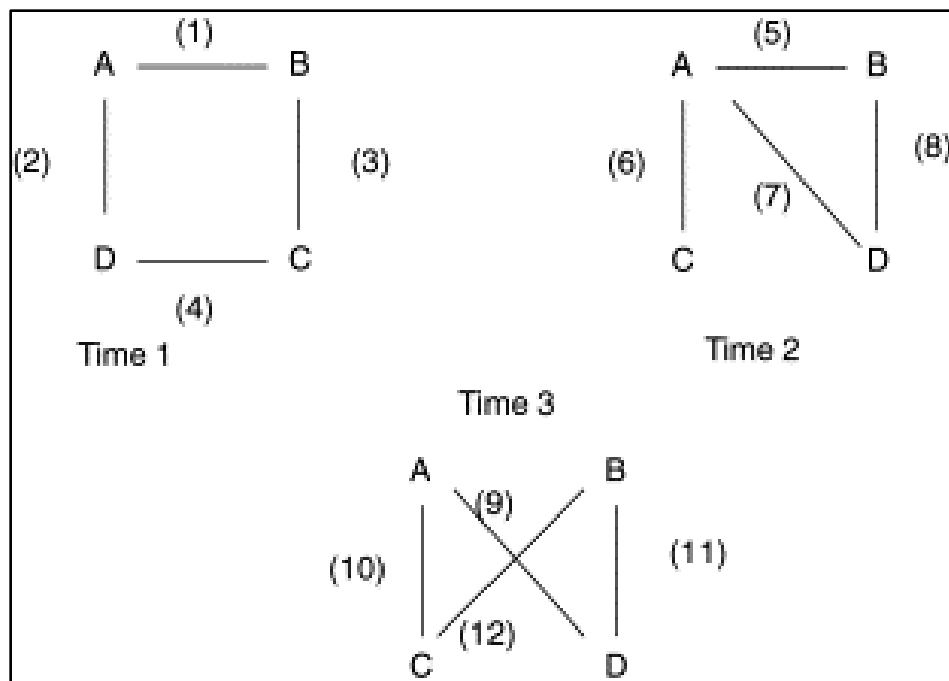


Hyperedges: Another way to represent data in “network” form



→ $E =$

A	B	C	D	t1	t2	t3
1	1	0	0	1	0	0
1	0	0	1	1	0	0
0	1	1	0	1	0	0
0	0	1	1	1	0	0
1	1	0	0	0	1	0
1	0	1	0	0	1	0
1	0	0	1	0	1	0
0	1	0	1	0	1	0
1	0	0	1	0	0	1
1	0	1	0	0	0	1
0	1	0	1	0	0	1
0	1	1	0	0	0	1

What is a clique?

- Let us now turn to the concept of a clique. In a very intuitive way, when we use the term sociometric clique, we mean a highly cohesive subgroup of individuals. Cohesiveness, however, has two possible dimensions.
 - In one dimension, which we will call **completeness**, a cohesive subgroup would be one in which a high proportion of its pairs possess the appropriate relation.
 - In the other, which we will call the **centripetal-centrifugal** dimension, a cohesive subgroup would be one in which relations among members of the subgroup are more important or more numerous than relations between members and non-members.
- We will return later to a formalization of these two dimensions. It is interesting to note that many informal definitions of cliques (Spilerman, 1966; Abelson, 1966) have referred solely to the centripetal-centrifugal dimension of cohesiveness, while Luce and Perry (1949) present a formalization of the completeness dimension of cohesiveness (Alba 1973: 115-116).

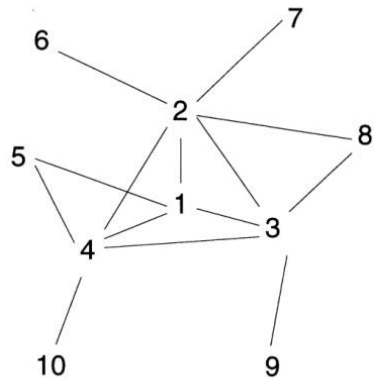
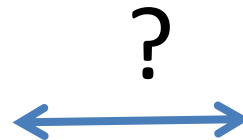


Fig. 1. A network with a core/periphery structure.



	1	2	3	4	5	6	7	8	9	10
1		1	1	1	1	0	0	0	0	0
2	1		1	1	0	1	1	1	0	0
3	1	1		1	0	0	0	1	1	0
4	1	1	1		1	0	0	0	0	1
5	1	0	0	1		0	0	0	0	0
6	0	1	0	0	0		0	0	0	0
7	0	1	0	0	0	0		0	0	0
8	0	1	1	0	0	0	0		0	0
9	0	0	1	0	0	0	0	0		0
10	0	0	0	1	0	0	0	0	0	

Observed Adjacency Matrix



	1	2	3	4	5	6	7	8	9	10
1		1	1	1	1	1	1	1	1	1
2	1		1	1	1	1	1	1	1	1
3	1	1		1	1	1	1	1	1	1
4	1	1	1		1	1	1	1	1	1
5	1	1	1	1		0	0	0	0	0
6	1	1	1	1	0		0	0	0	0
7	1	1	1	1	0	0		0	0	0
8	1	1	1	1	0	0	0		0	0
9	1	1	1	1	0	0	0	0		0
10	1	1	1	1	0	0	0	0	0	

Ideal Adjacency Matrix
Exhibiting Core/Periphery Structure

$$\rho = \sum_{i,j} a_{ij} \delta_{ij} \quad (1)$$

$$\delta_{ij} = \begin{cases} 1 & \text{if } c_i = \text{CORE or } c_j = \text{CORE} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

$$\delta_{ij} = \begin{cases} 1 & \text{if } c_i = \text{CORE and } c_j = \text{CORE} \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

$$\delta_{ij} = \begin{cases} 1 & \text{if } c_i = \text{CORE and } c_j = \text{CORE} \\ 0 & \text{if } c_i = \text{PERIPHERY and } c_j = \text{PERIPHERY} \\ . & \text{otherwise} \end{cases} \quad (4)$$